JHURALD HILARY A. LANTAPE

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SUMMARY

Data Engineer | Electronics Engineering (ECE) Student at the University of Southeastern Philippines (USeP). Specializing in decoupled backend architecture, event-driven ETL pipelines, and real-time operational intelligence. Expert in replacing fragmented manual workflows with secure, automated cloud systems that maintain strict data integrity.

TECHNICAL SKILLS

- **Backend & Data:** Python (Adv), FastAPI, PostgreSQL, Supabase, ETL, Idempotent Logic, Pydantic, Pandas.
- **Cloud & Data Engineering:** Azure Blob Storage, MotherDuck (DuckDB), Kestra, dbt (Core), Terraform, Parquet.
- **Automation & Web Scraping:** Browser Automation, Anti-Bot Mitigation, API Reverse-Engineering, Playwright, Requests.
- **Engineering & Tools:** Git/GitHub, LoRa, ESP32, SQLite, Regex, Excel (Adv), Video/Audio Transcoding.

CERTIFICATIONS

Data Engineering Zoomcamp 2026 | DataTalks.Club | April 2026

- Intensive 9-week program covering the Modern Data Stack: Docker, Terraform, Airflow, dbt, BigQuery, and Spark. Passed Final Capstone project.

TECHNICAL PROJECTS

Mathematics Research Data Pipeline — DE Zoomcamp Capstone | Azure, MotherDuck, Kestra, dbt, Terraform

- Architected an end-to-end batch pipeline ingesting 500k+ arXiv research papers (1992–Present) into Azure Data Lake using Terraform IaC; Kestra DAG chunks records into 10,000-row compressed Parquet batches, eliminating OOM crashes across the full historical backfill.
- Built a dbt dimensional model with idempotent MD5 surrogate keys ensuring 100% pipeline idempotency across reruns; optimized MotherDuck (DuckDB) query performance via native clustering on published_timestamp and primary_category, reducing dashboard load times significantly.
- Deployed a Metabase dashboard (custom Ubuntu Dockerfile for DuckDB JDBC compatibility) visualizing exponential paper growth over 3 decades, top mathematical sub-disciplines, and total ingestion volume scorecard.

Restaurant OS: Event-Driven ETL Microservice (Proj 7) | FastAPI, PostgreSQL, Webhooks

- Architected a high-speed FastAPI microservice ingesting async JSON payloads from Toast POS & 7shifts; used BackgroundTasks to compute real-time KPIs (CPLH, Contribution Margins) without blocking upstream services — reducing manual reporting time to zero.
- Engineered idempotent UPSERT logic in PostgreSQL (Supabase) guaranteeing 100% data integrity during out-of-order webhook events across concurrent SaaS data streams. Defined complex schema mappings and synchronization rules to normalize fragmented POS data into a unified reporting layer.

Cloud-Native Report Engine (Proj 6) | WordPress REST API Pipeline (Proj 5) | Python, Jinja2, WeasyPrint, WordPress REST API

- Designed a decoupled pipeline processing raw payloads through a custom diagnostic rules engine; automated branded PDF report generation via WeasyPrint with a No-Disk ingestion layer, eliminating manual report creation entirely.
- Built a production-grade ETL pipeline bridging raw data scraping to structured CMS ingestion via REST API mapping; applied MD5 fingerprinting to ensure full idempotency during high-volume reruns.

Enterprise Data Acquisition & IoT Suite (Proj 1–4) | LoRa, ESP32, Playwright, Pandas

- **Smart Park & LoRa Gateway:** Engineered dual-sensor fusion (Magnetometer + Ultrasonic) on ESP32 for real-time vehicle occupancy detection; built a LoRa-to-Cloud gateway streaming live data to Blynk IoT Cloud for remote monitoring.
- **API Extractors:** Developed an anti-bot mitigation pipeline bypassing enterprise WAFs (Incapsula) via session bridging, and a Chrome DevTools Protocol (CDP) session interceptor for high-fidelity Shopee market intelligence.

Intelligent Directory ETL Pipeline — Proj 9 | Playwright, Gemini API, FastAPI, WordPress REST API

- Architected an end-to-end AI-enriched ETL pipeline scraping business data (name, address, hi-res photos) from Google Maps via Playwright; leveraged Gemini API for LLM description generation and Vision-based image quality gates, discarding irrelevant assets before ingestion.
- Engineered idempotent WordPress REST API ingestion using MD5(name+address) as a surrogate place_id to prevent duplicate CPT records; built intelligent fallback logic routing to original scraped assets on AI quota failures, ensuring zero pipeline breakage. Validated end-to-end flow via a lightweight FastAPI mock WordPress server.

EDUCATION

Bachelor of Science: Electrical, Electronics and Communications Engineering (Expected Jun 2027)

University of Southeastern Philippines — Obrero, Davao City

- **Leadership:** Technical Head, AECES (2025–2026) — overseeing technical initiatives for the engineering student body.
- **Relevant Coursework:** Systems Logic, Advanced Mathematics, Technical Troubleshooting.

OTHER PROFESSIONAL EXPERIENCE

Digital Asset & Video Editor (Volunteer) | Global Impact, Davao City | Apr 2021 – Dec 2021

- Managed high-volume digital asset workloads for a global organization — transcoding video/audio files, maintaining organized backups in a DAM system, and delivering production-ready media assets on tight deadlines.